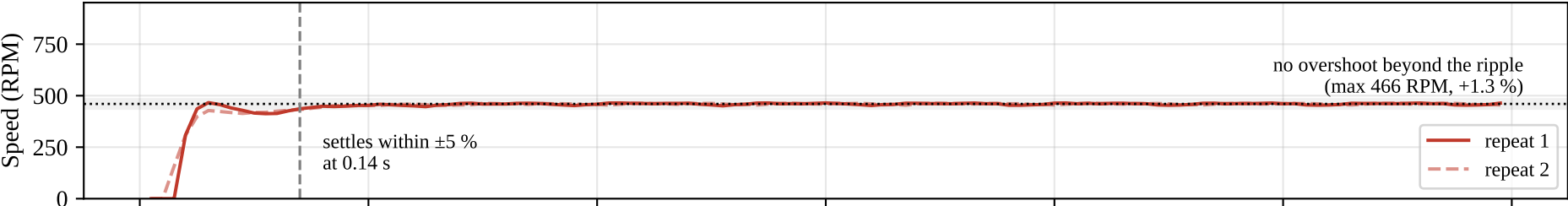
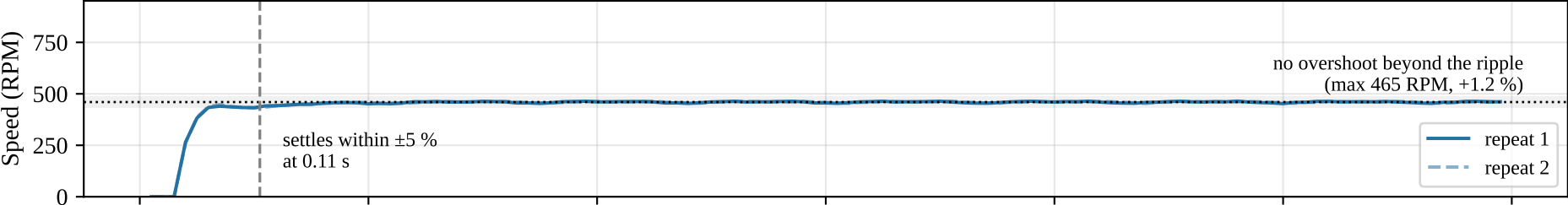


Step response 0 → 460 RPM, one panel per setting (solid = repeat 1, dashed = repeat 2; grey band = ±5 %)

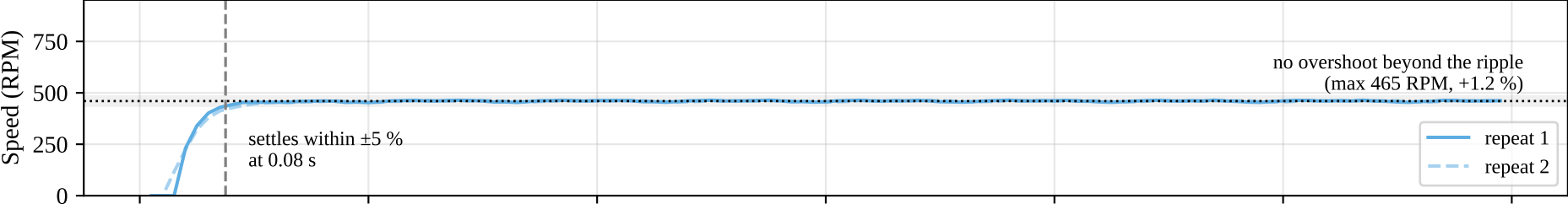
A: baseline (Kp 1.56, Ki 41.2) → overshoot 1.3 %, settles in 0.14 s



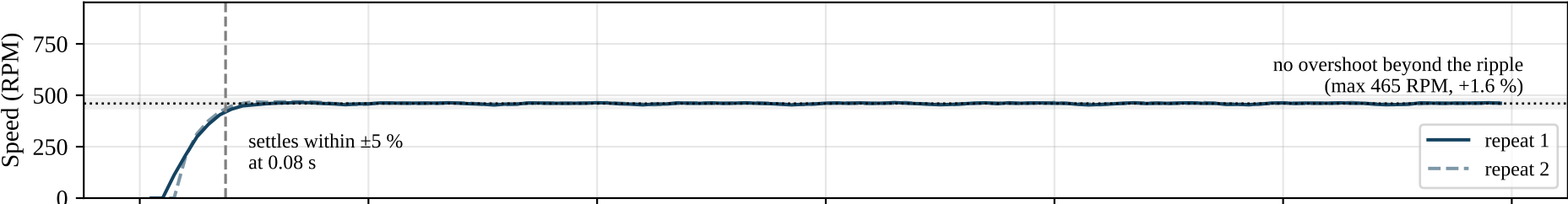
C1: Kp 1.2 → overshoot 1.2 %, settles in 0.11 s



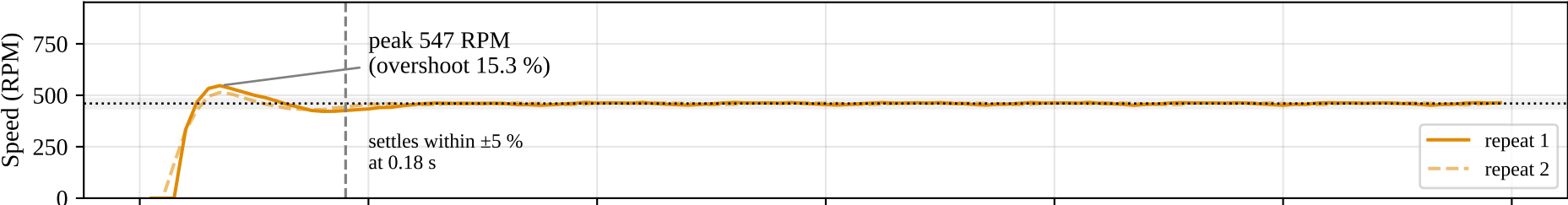
C2: Kp 1.0 → overshoot 1.2 %, settles in 0.08 s



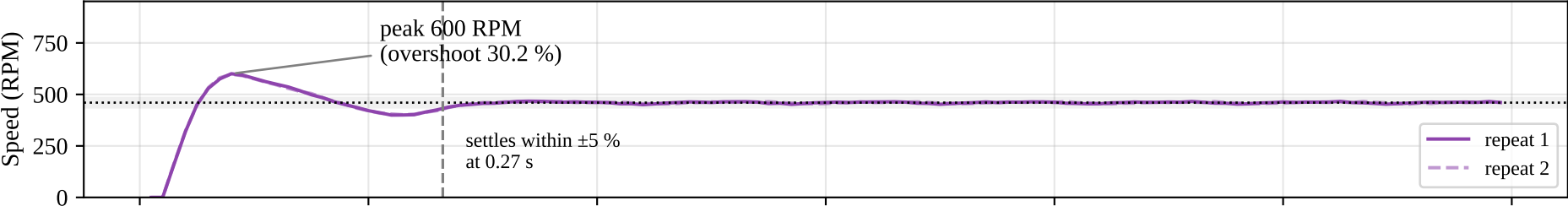
C3: Kp 0.8 → overshoot 1.6 %, settles in 0.08 s



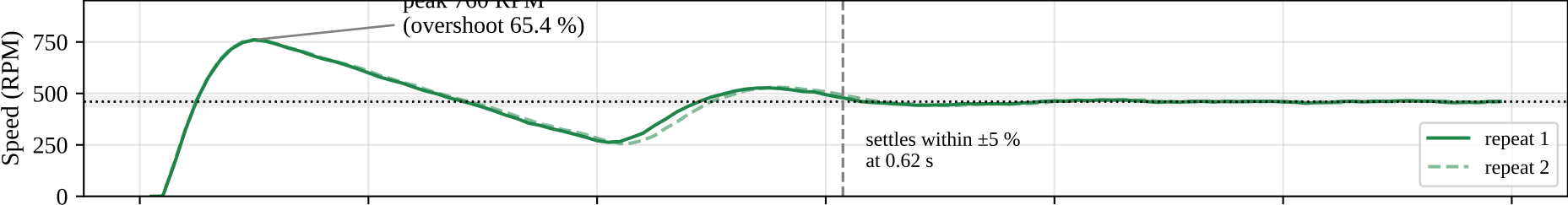
B1: α 0.5 → overshoot 15.3 %, settles in 0.18 s



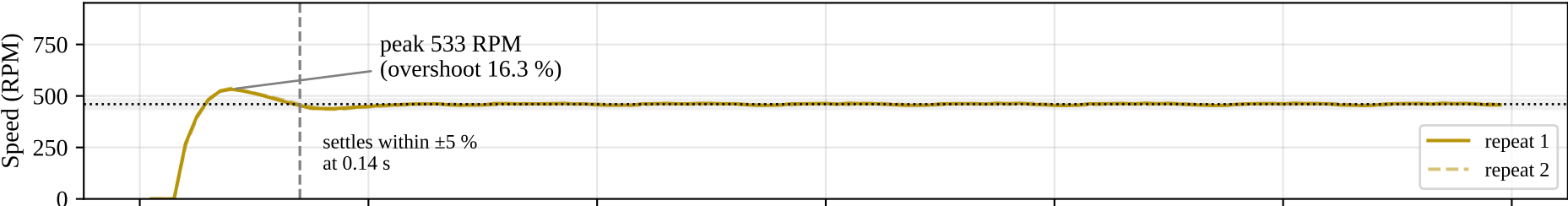
B2: α 0.3 → overshoot 30.2 %, settles in 0.27 s



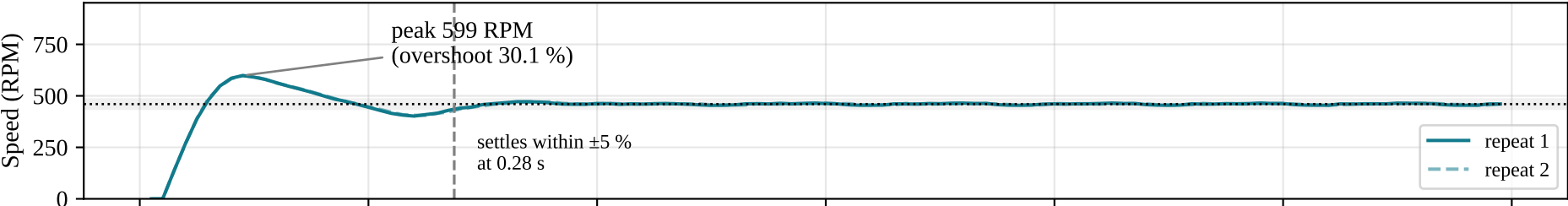
B3: α 0.15 → overshoot 65.4 %, settles in 0.62 s



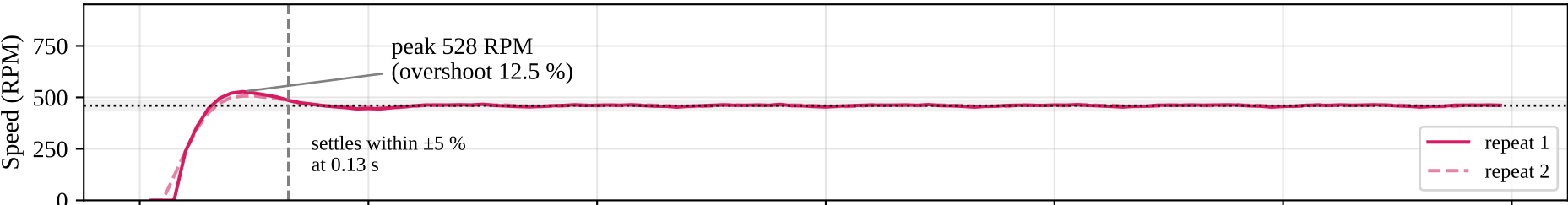
D1: Kp 1.2, α 0.5 → overshoot 16.3 %, settles in 0.14 s



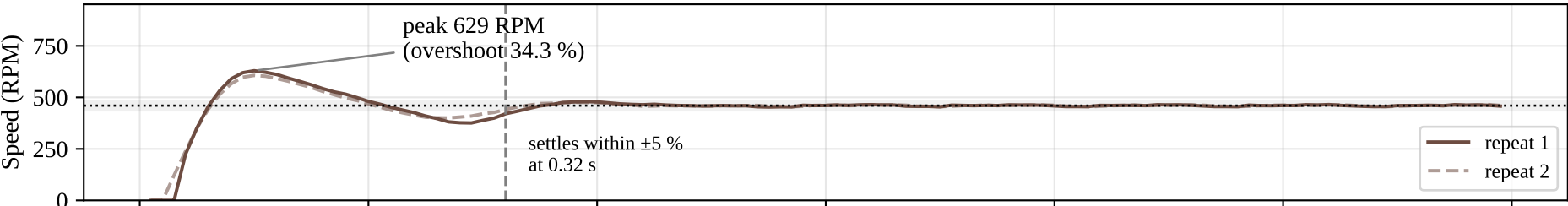
D2: Kp 1.2, α 0.3 → overshoot 30.1 %, settles in 0.28 s



D3: Kp 1.0, α 0.5 → overshoot 12.5 %, settles in 0.13 s



D4: Kp 1.0, α 0.3 → overshoot 34.3 %, settles in 0.32 s



Time (s)